



Fig. 6: Hybrid emulation enables early system-level hardware/software co-verification

capacity by moving processor models to the virtual prototype.

Since the advent of transaction-level modelling, it has been possible to create a virtual platform of a CPU sub-system, which trades off accuracy for speed in order to provide an early target to test software. Traditionally, the make or break of such virtual platforms would depend on the availability of SystemC models for various components; it would simply take too long to generate a trustworthy model for, say, a new coprocessor, so the benefit of early software simulation would be lost. Growth of SystemC model libraries for popular functions such as ARM's Fast Models has helped fill those gaps, but what about the new functions unique to the new SoC?

Designers and commentators often use the term software freely, but without specifically defining what they mean. An SoC, like most other embedded processing systems, has different kinds of software, operating and inter-reacting at different levels of a software stack and across multiple processors. User space (Fig. 7) including applications appears at the top of the stack and rests upon lower layers, which are increasingly concerned with the operating system kernel and the hardware as you reach lower levels.

Choose any level in the stack. If you wish to develop software at that level, then you need a representation of all levels below upon which to run it. Ideally, that might be the real lower-level software running in real system hardware; but when that

is not available, a model must be created instead. Well, it should have just enough accuracy at a given level in order to maintain the deception that the software at that level is running in the real system; any accuracy greater than that will just be a waste of simulation time. Software at the lowest levels of the stack is aware of SoC hardware and masks any hardware dependencies from higher-level software, thereby allowing greater portability and reuse of programs in the user space.

Software at these lower levels is called hardware-dependent software. A model on which you can test aspects of hardware-dependent software, such as a BSP or BIOS, needs high accuracy, and may include a cycle-accurate model of the relevant hardware itself. This may be in an RTL simulator, emulator or FPGA based prototype, depending on how fast you need to run. Software at the highest levels of the stack, such as apps and other user space programs, need the least accuracy, and can therefore run at the highest speeds.

Virtual platforms and FPGA based emulation

The task of a virtual platform is to have just enough accuracy to support the level of software being run on it. This is largely achieved by modelling the behaviour and inter-block communications at transaction level, which makes these inherently faster than equivalent cycle-accurate representations. If you have the necessary models or libraries of models usually written in SystemC,

INSULATION TESTERS

Analog

MC-9008A Series

Model	Range	Test Voltage DC
MC908A	0-20M Ohms	100V
MC908A	0-100M Ohms	300V
MC908A	0-500M Ohms	500V
MC908A	0-1000M Ohms	500V
MC908A	0-500M Ohms	1000V
MC908A	0-500M Ohms	1000V
MC908A	0-2000M Ohms	1000V

Digital

DIT 99 Series

Model	Range	Test Voltage DC	Resolution
DIT 99A	0-20M Ohms	100V	0.01M Ohms
DIT 99B	0-200M Ohms	250V	0.1M Ohms
DIT 99C	0-200M Ohms	500V	0.1M Ohms
DIT 99D	0-200M Ohms	1000V	0.1M Ohms
DIT 99E	0-2000M Ohms	1000V	1M Ohms

5kV

DIT 954

Specification	Test Voltage	Range
Insulation Resistance	1000V / 2500V / 5000V	0.1M to 2000M
AC Voltage Measurement	0-600V AC (50-60Hz)	-
Phase Sequence Test	100V-450V (Phase-Phase 40-40Hz)	-

Meco Meters Private Limited

Plot No. EL-50, MIDC Electronic Zone,
TTC Industrial Area, Malape,
New Mumbai - 400110 (MCA)

Tel : 0091-22-27673300 (Board), 27673311-16 (Sales)
Fax : 0091-22-27673310, 27673330
Email : sales@meconist.com
Web : www.meconist.com